

EMMANUEL CHERUTICH

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PROFESSIONAL-SUMMARY

Detail-oriented and innovative person with hands-on experience in AI, machine learning, software engineering, automation, and intelligent systems. Passionate about leveraging technology to solve real-world problems, empower communities, and build scalable solutions. Proven ability in building end-to-end production systems, AI-powered chatbots, financial applications, and automation tools. Experienced technical mentor and trainer with a strong commitment to tech education and community growth.

EDUCATION

Bachelor of Science in Computer Science

Technical University of Mombasa

Key Coursework: Machine Learning, Data Structures, Algorithms, Cybersecurity, Web Development, Distributed Systems

TECHNICAL SKILLS

Languages: Python, JavaScript, SQL, HTML/CSS

Frameworks & Libraries: Flask, FastAPI, TensorFlow, Keras, PyTorch, Pandas, NLTK, Hugging Face, Kivy, React, Lang Chain

Tools & Platforms: Ubuntu Linux, Anaconda, Git, VS Code, MySQL/phpMyAdmin, PythonAnywhere, Chroma DB

AI & ML: LSTM, ARIMA, TF-IDF, Cosine Similarity, Character-level NLP, Speech Transcription, Computer Vision, Face Recognition, Fine-tuning LLMs

APIs & Integrations: WhatsApp Business API, Twilio, M-Pesa STK Push, M-Pesa Daraja, PayPal, Bank APIs

PROJECTS

Ewe Language Speech Transcription Model

Built and trained a custom speech-to-text model for transcribing the Ewe language, addressing low-resource language gaps using PyTorch and fine-tuned Hugging Face models.

AI-Powered WhatsApp Bot

Developed a fully automated WhatsApp bot using WhatsApp Business API, capable of intelligent query processing, AI model integration, and safe communication outside group chats.

Website AI Chatbot (PDF-powered RAG System)

Created an intelligent chatbot using RAG (Retrieval-Augmented Generation) to answer user queries by retrieving information from internal PDF documents.

Investment Group Finance System

Created an end-to-end financial management system for an investment group, including member accounts, monthly contributions, automated statements, withdrawals, and admin analytics.

Referral-Based M-Pesa Payment System

Developed an automated M-Pesa STK Push system with referral functionality, instant user rewards, dashboards, authentication, and live transaction tracking.

Timetable Simplifier for Technical University of Mombasa

Created a web-based system that processes complex Excel timetables, identifies courses and years, and simplifies PDF timetables for designated classes.

AI-Powered Cultural Community Platform

Developed a voice and text-based community platform that fetches data from a vector database to facilitate cultural activities.

Face Recognition & Fruit Classification Models

Developed a face recognition model for a Face ID system and a computer vision model for fruit classification.

EXPERIENCE

Technical Mentor

Technical University of Mombasa (2024 – Present)

Mentored students in AI, ML, Python, and Linux

Organized study groups, AI workshops, and hackathon teams

Supported project development, model debugging, and application deployment

Software Development Intern

Swahilipot Hub Foundation (May 2025 – Aug 2025)

Supported web and backend development projects

Developed APIs, optimized database structures, and assisted digital community projects

Worked in an agile team environment

Frontend Technical Trainer

Swahilipot Hub (June 2025 – Oct 2025)

Trained students in HTML, CSS, JavaScript, and UI/UX basics

Designed practical projects and training sessions to develop youth tech skills

Conducted weekly classes and mentorship initiatives

ACHIEVEMENTS

- Developed and deployed several AI applications in real-world settings
- Mentored scores of students in AI, programming, and software development
- Finalist/Participant in various hackathons and local tech competitions
- Developed a mobile app for a secretary at CIG to track expenses efficiently

INTERESTS

AI & Cybersecurity, Robotics & Automation, Time Series Forecasting, Community Tech Education.